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**Can hedging tell the full story? Reconciling differences in United States
 aggregate- and industry-level exchange rate risk premium**

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1 Big picture

- **Question.** Why does prior work find that currency risk is important for the aggregate U.S. stock market but not for U.S. industries? Is the missing industry-level currency premium due to firms hedging away the risk, or due to weak empirical methods?
- **Main answer.** Methodology explains the puzzle better than hedging. Once currency risk is modeled conditionally with time-varying betas and time-varying expected factor returns, all 36 U.S. industries have significant currency risk premiums.
- **Empirical object.** The paper studies the *ex ante currency risk premium* in industry expected returns, not only the ex post sensitivity of realized returns to realized exchange-rate changes.
- **Risk factors.** The asset-pricing model has five factors: the Fama–French market, SMB, and HML factors, plus two exchange-rate factors: MAJOR currencies from developed trading partners and OITP currencies from other important trading partners, mostly emerging economies.
- **Conditional model.** Expected industry returns are modeled as time-varying betas multiplied by conditionally expected factor returns. The time-varying covariances and variances come from a multivariate GARCH specification.
- **Core result.** The average absolute currency risk premium is about 2.47 percentage points per year and accounts for about 11.7% of total industry risk premium.
- **Economic result.** Currency premia are sometimes positive and sometimes negative, so currency risk can raise or lower the cost of equity depending on the industry and period.
- **Cross-sectional determinants.** Foreign income, industry concentration, leverage, liquidity, firm size, market-to-book, and nontraded-industry dummies explain meaningful cross-industry variation in currency premia.
- **Time-series determinants.** Foreign trade, macro growth opportunities, monetary policy, recession periods, and currency-crisis dummies explain time variation in industry currency premia.
- **Takeaway.** Hedging does not tell the full story. Exchange-rate risk remains priced in U.S. industry expected returns when the model allows exposure and risk premia to vary through time.

2 Data and measurement

2.1 Industry portfolios

- The sample covers monthly data from January 1980 to December 1999.
- The paper uses 36 U.S. industry portfolios based on Fama and French value-weighted industry portfolios.
- The industries include 31 traded-goods/manufacturing industries and five nontraded-goods industries: entertainment, construction, retail, restaurants/hotels/motels, and banking.
- Industry returns are measured in excess of the risk-free rate.

Interpretation: The industry, not the firm or aggregate market, is the unit of analysis because exchange-rate exposure can differ greatly across industries even inside one country.

2.2 Equity risk factors

The model includes the three Fama–French equity factors:

- r_{Vt} : the value-weighted U.S. market excess return;
- r_{St} : SMB, the size factor;
- r_{Ht} : HML, the book-to-market factor.

Interpretation: Including SMB and HML reduces the chance that the estimated currency premium is merely proxying for omitted domestic equity risk.

2.3 Currency risk factors

The two exchange-rate factors are:

- f_{Mt} : MAJOR, percentage changes in the real U.S. Treasury trade-weighted exchange-rate index for 16 developed-country trading partners;
- f_{Ot} : OITP, percentage changes in the real index for the currencies of other important trading partners, mainly from developing economies.

The indices are expressed as foreign currency per U.S. dollar. Therefore, a positive percentage change is an appreciation of the dollar.

Interpretation: Splitting MAJOR and OITP is important for the hedging hypothesis. Developed-country currency risk is easier to hedge, while emerging-market currency risk is harder to hedge. If hedging fully explained the weak prior results, the OITP premium should still be important even if the MAJOR premium disappeared.

2.4 Instrumental variables

Expected factor returns are estimated using lagged instruments. For equity factors, the instruments include:

- default premium;
- term premium;
- Federal funds rate;

- lagged dependent variable.

For currency factors, the instruments include:

- U.S. export ratio: aggregate exports as a percentage of GDP;
- U.S. import ratio: aggregate imports as a percentage of GDP;
- Federal funds rate;
- lagged dependent variable.

Interpretation: The conditional model relies on the idea that investors forecast both equity factor returns and exchange-rate changes. The instruments are used to construct those conditional expectations.

2.5 Industry-level characteristics

The cross-sectional tests use industry characteristics such as:

- foreign income divided by total sales;
- acid ratio as a liquidity measure;
- Herfindahl index of industry sales;
- market-to-book ratio;
- log market equity;
- leverage;
- nontraded-industry dummies.

Interpretation: These variables ask whether currency premia are larger in industries where exchange-rate risk should plausibly matter more: internationally exposed, competitive, levered, or liquidity-constrained industries.

2.6 Macro variables for time-series tests

The time-variation regressions include:

- NBER recession dummy;
- Mexican currency crisis dummy;
- Asian currency crisis dummy;
- Volcker tight-money experiment dummy;
- export ratio and import ratio;
- growth opportunities, measured using the S&P 500 price-earnings ratio.

Interpretation: These variables are used as a model-specification check. If estimated currency premia are economically meaningful, they should vary with macro conditions that affect currency risk and trade competitiveness.

3 Models and empirical specifications

3.1 Conditional expected return equation

The main asset-pricing equation is

$$E_{t-1}(r_{it}) = \beta_{iV,t-1}E_{t-1}(r_{Vt}) + \beta_{iS,t-1}E_{t-1}(r_{St}) + \beta_{iH,t-1}E_{t-1}(r_{Ht}) \quad (1)$$

$$+ \beta_{iO,t-1}E_{t-1}(f_{Ot}) + \beta_{iM,t-1}E_{t-1}(f_{Mt}). \quad (2)$$

Here r_{it} is the excess return on industry i , V, S, H denote market, SMB, and HML, and O, M denote OITP and MAJOR exchange-rate factors.

Interpretation: Expected industry returns are the sum of risk quantities, the betas, times risk prices or expected factor returns. Currency risk is priced if the conditional expected currency factors contribute significantly to expected industry returns.

3.2 Realized return equation

The realized industry return equation is

$$r_{it} = \beta_{iV,t-1}(r_{Vt} - \varepsilon_{Vt}) + \beta_{iS,t-1}(r_{St} - \varepsilon_{St}) + \beta_{iH,t-1}(r_{Ht} - \varepsilon_{Ht}) \quad (3)$$

$$+ \beta_{iO,t-1}(f_{Ot} - \varepsilon_{Ot}) + \beta_{iM,t-1}(f_{Mt} - \varepsilon_{Mt}) + \varepsilon_{it}. \quad (4)$$

Interpretation: The observed factor realization is decomposed into a conditional expectation plus a residual. The model uses conditional expectations, not just realized exchange-rate changes, which is the key methodological difference from much of the earlier exposure literature.

3.3 Expected factor returns

The equity factor expectations are estimated as

$$r_{Vt} = a_0 + a_1 z_{1,t-1} + \cdots + a_4 z_{4,t-1} + \varepsilon_{Vt}, \quad (5)$$

$$r_{St} = b_0 + b_1 z_{1,t-1} + \cdots + b_4 z_{4,t-1} + \varepsilon_{St}, \quad (6)$$

$$r_{Ht} = c_0 + c_1 z_{1,t-1} + \cdots + c_4 z_{4,t-1} + \varepsilon_{Ht}. \quad (7)$$

The currency factor expectations are estimated as

$$f_{Ot} = d_0 + d_1 x_{1,t-1} + \cdots + d_4 x_{4,t-1} + \varepsilon_{Ot}, \quad (8)$$

$$f_{Mt} = g_0 + g_1 x_{1,t-1} + \cdots + g_4 x_{4,t-1} + \varepsilon_{Mt}. \quad (9)$$

Interpretation: The instruments allow factor risk premia to vary through time. This is crucial because constant-parameter models can understate the economic role of exchange-rate risk.

3.4 Time-varying betas

The factor beta is defined using conditional second moments:

$$\beta_{ik,t-1} = \frac{\text{cov}_{t-1}(r_{it}, r_{kt})}{\text{var}_{t-1}(r_{kt})}. \quad (10)$$

For the currency factors, replace r_{kt} with f_{Ot} or f_{Mt} .

Interpretation: Currency exposure is not a fixed coefficient. It changes as the covariance between industry returns and currency factors changes, and as exchange-rate volatility changes.

3.5 GARCH covariance system

Let

$$e_t = (\varepsilon_i, \varepsilon_V, \varepsilon_S, \varepsilon_H, \varepsilon_O, \varepsilon_M)'. \quad (11)$$

The conditional covariance matrix is modeled as

$$H_t = C'C + A_1'e_{t-1}e_{t-1}'A_1 + B_1'H_{t-1}B_1. \quad (12)$$

Interpretation: The multivariate GARCH model generates the time-varying variances and covariances needed to estimate the conditional betas.

3.6 Currency risk premium

For industry i , the OITP and MAJOR currency premia are

$$CRP_{it}^O = \beta_{iO,t-1}E_{t-1}(f_{Ot}), \quad (13)$$

$$CRP_{it}^M = \beta_{iM,t-1}E_{t-1}(f_{Mt}). \quad (14)$$

The total currency risk premium is

$$CRP_{it} = CRP_{it}^O + CRP_{it}^M. \quad (15)$$

Interpretation: The currency beta alone is not the risk premium. The premium depends on both exposure and the expected exchange-rate factor return.

3.7 Economic importance measure

The paper reports the absolute currency premium as a share of total absolute risk premium:

$$\text{Currency share}_i = \frac{|CRP_i^O| + |CRP_i^M|}{|RP_i^V| + |RP_i^S| + |RP_i^H| + |CRP_i^O| + |CRP_i^M|}. \quad (16)$$

Interpretation: This measure asks how much of the total expected-return compensation is attributable to currency risk, ignoring signs so that positive and negative premia do not cancel.

3.8 Cross-sectional determinant regression

The cross-sectional tests estimate regressions of the form

$$CRP_{it}^j = \alpha + \Gamma'X_{it} + \Lambda'IndustryDummies_i + \delta_t + u_{it}, \quad j \in \{O, M\}, \quad (17)$$

where X_{it} contains foreign income, liquidity, concentration, market-to-book, size, and leverage.

Interpretation: If the estimated currency premia are economically meaningful, industry characteristics that affect currency exposure or hedging ability should explain them.

3.9 Time-series determinant regression

For each industry, the time-series tests estimate a regression like

$$CRP_{it}^j = \alpha_i + \Theta_i' Z_t + \eta_{it}, \quad (18)$$

where Z_t includes recession, currency-crisis, trade, monetary-policy, and growth-opportunity variables. The paper reports sums of contemporaneous and lagged coefficients.

Interpretation: This test is both substantive and diagnostic. Currency premia should move with trade, macro conditions, and crisis episodes if they are not merely statistical artifacts.

4 Empirical design and interpretation

4.1 What makes the exercise different from older exposure regressions

- Older work often regresses realized industry returns on realized exchange-rate changes using constant coefficients.
- This paper studies conditional expected returns and conditional currency premia.
- Both exposures and expected factor returns are allowed to change over time.
- The model separates developed-market currency risk from emerging-market currency risk.

Interpretation: The paper's claim is not that earlier studies were numerically wrong. The claim is that their unconditional design was too restrictive to detect a priced, time-varying currency premium.

4.2 Hedging hypothesis versus methodology hypothesis

- **Hedging hypothesis:** If firms hedge currency risk effectively, currency premia should be small or absent, especially for easily hedged developed-market currencies.
- **Methodology hypothesis:** If previous papers used overly restrictive constant-parameter models, a conditional model should reveal significant premia.
- The paper finds significant premia for both OITP and MAJOR factors.

Interpretation: The evidence favors the methodology hypothesis. If hedging fully eliminated the premium, the MAJOR factor should be weak, but it is not.

4.3 Why industry analysis matters

- Currency movements affect competitiveness differently across industries.
- Some industries have foreign sales; others face import competition or rely on foreign inputs.
- Nontraded industries can still be indirectly exposed through aggregate demand, capital reallocation, or links to traded sectors.

Interpretation: Aggregate stock-market evidence cannot tell which industries bear currency risk or whether currency risk can be diversified across domestic industries.

4.4 Limitations

- The paper uses industry portfolios, so it does not observe each firm's hedging contracts directly.
- The model is estimated one industry at a time, which sacrifices some cross-industry efficiency.
- The currency factors are indices, not all bilateral exchange rates separately.
- The model relies on instrumented expected factor returns and GARCH-based covariance estimates, so interpretation depends on the conditional asset-pricing specification.

Interpretation: These limitations do not overturn the paper's main point, but they matter for interpreting the results as evidence of priced exchange-rate risk rather than direct firm-level operating exposure.

5 Tables and figures

This section keeps the *Read* plus *Economic message* style and groups adjacent original tables/figures to save space. The images are cropped directly from the paper.

5.1 Table 1: industry and factor summary statistics

Read:

- The data include 36 industry portfolios from January 1980 to December 1999.
- Industry mean excess returns differ greatly: coal has a negative mean, while tobacco, retail, entertainment, drugs, and other industries have high mean excess returns.
- Panel B reports the market, SMB, HML, OITP, and MAJOR factor moments.
- The currency factors have low average returns but meaningful volatility and autocorrelation.

Economic message: The table establishes that industries differ substantially in risk and return, and that the exchange-rate factors contain time-series variation that can potentially be priced.

Table 1
Summary statistics of returns on industries and risk factors

Panel A reports summary statistics for 31 manufacturing industries and five nontraded goods industries. These are from the Fama and French (1997) value-weighted industry portfolios. Returns (percent per month) are in excess of the risk-free rate, which has a mean of 0.559% per month. ρ_1 is the first-order autocorrelation. J-B is the Jarque-Bera statistic for the test of the null hypothesis that the returns are normally distributed. Nontraded goods industries, using the definitions in Bodnar and Gentry (1995), have (n) to the right of their name. In Panel B, the risk factors are the Fama and French (1993) factors: market excess return on the Center for Research in Security Prices value-weighted market (VWM), returns on small minus big firms (SMB), and returns on high minus low book-to-market value firms (HML). The currency factors are represented by percentage changes in the real US Treasury trade-weighted exchange rate index made up of the currencies of 16 developed countries that are the main trading partners of the US (MAJOR) and the real index of the currencies of the other important trading partners (OITP) from the emerging economies. The indices are in foreign currency per US dollar. Numbers in boldface are significant at least at the 5% significance level.

Panel A: Summary statistics of portfolio excess returns

Industry	Mean	Standard deviation	ρ_1	J-B
Agri	0.690	6.161	-0.005	652.8
Food	0.915	4.739	0.011	16.59
Soda	1.018	5.128	-0.046	29.14
Beer	1.108	6.503	0.119	26.48
Smoke	1.319	7.910	0.059	23.78
Toys	0.989	6.729	0.047	7.833
Fun (n)	1.285	5.647	0.014	91.65
Books	1.029	5.469	0.138	53.88
Holdd	1.095	4.783	0.007	52.47
Cltbs	0.488	6.178	0.242	126.5
MedEq	1.004	5.604	0.017	7.016
Drugs	1.115	5.175	-0.001	10.35
Chemls	0.772	5.416	0.015	149.4
Rubbr	0.728	5.879	0.065	195.3
Textls	0.580	6.056	0.264	222.8
Blldtlt	0.709	5.622	0.106	147.8
Cnst (n)	0.246	7.153	0.236	44.28
Steel	0.430	6.458	-0.025	126.6
FabPr	0.336	6.230	0.160	66.77
Mach	0.434	5.951	0.122	140.6
ElcEq	1.053	6.606	0.071	392.2
Autos	0.802	5.931	0.124	71.13
Aero	0.871	6.356	0.091	106.9
Ships	0.621	6.819	0.074	46.15
Guns	0.670	6.811	0.083	138.6
Gold	0.299	11.11	0.024	34.27
Mines	0.135	7.038	0.004	67.60
Coal	-0.116	6.510	0.029	49.81
Compus	0.852	6.317	0.098	3.130
Chips	1.154	6.827	0.056	9.610
LabEq	1.067	8.001	0.042	2.340
Paper	0.759	5.827	-0.012	99.02
Boxes	0.966	5.694	-0.023	284.5
Rtail (n)	1.203	5.729	0.194	84.81
Meals (n)	0.761	5.520	0.164	42.41
Bankn (n)	0.969	5.958	0.156	51.11

Panel B: Summary statistics of returns on the risk factors

Risk factors	Mean	Standard deviation	ρ_1 (first-order autocorrelation)	J-B statistic
VWM (market)	0.854	4.401	0.051	193.310
SMB (size)	-0.029	2.666	0.132	4.367
HML (book-to-market)	0.188	2.799	0.225	2.725
OITP currency index	0.110	1.229	0.203	280.642
MAJOR currency index	0.036	1.830	0.318	1.618

industries are exposed to the respective indices. Several other specifications led to the same qualitative results.

3.3. Information instruments

Table 1: Summary statistics of returns on industries and risk factors

5.2 Tables 2–3: unconditional exposure benchmark and factor predictability

Read:

- Table 2 replicates the older style of OLS exposure regression: market-adjusted industry returns on realized currency-index changes.

- Only a minority of industries show significant ex post currency exposure, consistent with prior weak evidence.
- Table 3 shows that risk factors are predictable using lagged instruments; the null that all predictive coefficients are zero is rejected for several factors.
- Factor predictability is especially important because the paper prices currency risk through expected, not merely realized, exchange-rate changes.

Economic message: The unconditional OLS benchmark looks weak, but the conditional model has predictive information to work with. This contrast motivates the paper’s methodology.

Table 2

Estimates of exchange rate exposure

The table reports ordinary least squares betas (exposure) from regressing the market-adjusted returns of each industry on the changes in the real US Treasury trade-weighted exchange rate index made up of the currencies of 16 developed countries that are the main trading partners of the US (MAJOR) and the real index of the currencies of the other important trading partners (OITP) from the emerging economies. Following Griffin and Stulz (2001), the market return is subtracted from each industry’s return. Data are monthly from January 1980 to December 1999 (240 observations). Numbers in bold face are significant at least at the 5% significance level.

Industry	OITP coefficient	MAJOR coefficient	Adjusted R ²
Agri	−0.571	−0.120	1.689
Food	0.329	−0.048	0.298
Fun (n)	−0.191	0.087	−0.532
Soda	0.304	−0.066	0.033
Beer	0.149	0.204	−0.207
Smoke	−0.210	0.033	−0.641
Toys	0.000	0.401	1.644
Books	0.070	0.273	2.196
Hshld	0.152	0.125	0.920
Clths	−0.277	0.342	1.773
MedEq	0.256	−0.048	−0.118
Drugs	0.426	−0.147	1.514
Chem	0.093	−0.105	−0.363
Rubbr	−0.324	0.024	0.471
Textls	−0.401	0.254	1.114
BldMt	−0.152	−0.062	−0.019
Const (n)	−0.218	0.073	−0.435
Steel	−0.200	−0.108	−0.176
FabPr	−0.316	0.062	−0.046
Mach	−0.453	−0.088	3.003
ElcEq	−0.470	0.037	1.339
Autos	0.176	0.135	0.008
Aero	−0.107	0.184	−0.137
Ships	0.103	0.520	2.709
Guns	0.146	0.044	−0.661
Gold	−0.062	−1.022	2.560
Mines	−0.321	−0.328	1.484
Coal	0.018	0.081	−0.742
Comps	−0.155	−0.161	−0.007
Chips	−0.585	0.054	2.251
LabEq	−0.900	0.123	3.219
Paper	−0.089	−0.042	−0.650
Boxes	0.016	−0.107	−0.537
Rtail (n)	0.019	0.350	3.063
Meals (n)	−0.040	0.209	0.633
Banks (n)	0.065	0.352	2.664

Table 3

Predictability of risk factors

The table reports coefficients (and robust *p*-values) from regressions used to test if the factor returns are predictable. The instrumental variables are the default premium (DEFAULT), the spread between the yields on Moody’s Baa- and Aaa-rated corporate bonds; the term premium (TERM), the difference in the yields of the Treasury constant-maturity 10-year and the 1-year notes; the Fed funds rate (FED); and the lagged dependent variable (the risk factor). The currency factors are represented by percentage changes in the real US Treasury trade-weighted exchange rate index made up of the currencies of 16 developed countries that are the main trading partners of the US (MAJOR) and the real index of the currencies of the other important trading partners (OITP) from the emerging economies. The indices are in foreign currency per US dollar. For the currencies, the instruments are EXPRATIO, the percentage of US exports to gross domestic product (GDP), IMPRATIO, the percentage of US imports to GDP, FED, and the lagged dependent variable (the risk factor). All instruments are lagged one period relative to the factor returns. Data are monthly from January 1980 to December 1999 (240 observations).

Dependent variable	Constant	DEFAULT _{<i>t</i>−1}	TERM _{<i>t</i>−1}	FED _{<i>t</i>−1}	Dep variable _{<i>t</i>−1}	H ₀ : all coeff = 0	Adjusted R ²
VWM (market)	3.288 (0.007)	3.708 (0.001)	−0.992 (0.022)	−0.734 (0.000)	−0.024 (0.749)	15.266 (0.004)	4.60
SMB (size)	−0.442 (0.569)	2.071 (0.000)	−0.118 (0.664)	−0.2340 (0.035)	0.075 (0.338)	21.812 (0.000)	5.52
HML (book-to-market)	−2.314 (0.004)	−2.005 (0.001)	0.960 (0.001)	0.489 (0.000)	0.154 (0.046)	20.547 (0.000)	9.77
Dependent variable	Constant	IMPRATIO _{<i>t</i>−1}	EXPRATIO _{<i>t</i>−1}	FED _{<i>t</i>−1}	Dep variable _{<i>t</i>−1}	H ₀ : all coeff = 0	Adjusted R ²
BROAD currency index	−0.182 (0.422)	0.005 (0.762)	−0.023 (0.085)	0.030 (0.329)	0.329 (0.000)	45.315 (0.000)	11.85
OITP currency index	−0.242 (0.245)	−0.021 (0.178)	−0.006 (0.634)	0.047 (0.051)	0.192 (0.118)	15.383 (0.004)	6.55
MAJOR currency index	−0.167 (0.553)	0.019 (0.347)	−0.032 (0.059)	0.026 (0.497)	0.320 (0.000)	44.379 (0.000)	10.12

Table 2: OLS estimates of exchange-rate exposure

Table 3: Predictability of risk factors

5.3 Tables 4–5 and Figure 1: time-varying betas and currency exposure

Read:

- Table 4 reports time-varying betas relative to market, SMB, and HML. The market beta is positive and significant for every industry.
- Table 5 reports OITP and MAJOR currency betas. Mean absolute OITP betas average about 0.492, while mean absolute MAJOR betas average about 0.246.
- Many industry currency betas are significant, and the sign of exposure varies across industries and over time.
- Figure 1 illustrates large time variation in currency betas for selected high- and low-impact industries.

Economic message: Currency exposure is not stable. It is time-varying and industry-specific, which is why constant-parameter regressions can miss economically important currency risk.

Table 4

Time-varying industry betas relative to the equity factors

The table reports the sample mean of the time-varying equity-related factor betas estimated from the system of Eqs. (2)–(9). The portfolios are 31 manufacturing industries and five nontraded goods industries. Nontraded goods industries, using the definitions in Bodnar and Gentry (1993), have (n) to the right of their name. The equity risk factors are the Fama and French (1993) factors: excess returns on the Center for Research in Security Prices value-weighted market index (VWM), returns on small minus big firms (SMB), and returns on high minus low book-to-market value firms (HML). Significance at the 1% level is represented by *.

Industry	VWM	SMB	HML
<i>Panel A: Mean of estimated time-varying betas</i>			
Agri	0.864*	0.696*	−0.693*
Food	0.794*	−0.309*	−0.373*
Soda	0.867*	−0.387	−0.679*
Beer	0.776*	−0.163*	−0.572*
Smoke	0.920*	−0.237*	−0.354*
Toys	1.258*	0.662*	−1.382*
Books	1.026*	0.184*	−0.620*
Fun (n)	1.121*	0.449*	−0.810*
Hshld	1.005*	−0.214*	−0.769*
Clths	1.109*	0.602*	−0.942*
MedEq	1.029*	0.037*	−1.092*
Drugs	0.968*	−0.344*	−0.899*
Chems	1.070*	0.089*	−0.695*
Rubbr	1.140*	0.751*	−0.974*
Txltls	0.968*	0.960*	−0.699*
BldMt	1.124*	0.250*	−0.758*
Cnst (n)	1.251*	0.801*	−0.905*
Steel	1.038*	0.559*	−0.640*
FabPr	0.930*	0.857*	−0.692*
Mach	1.158*	0.655*	−0.951*
ElcEq	1.269*	0.999*	−1.476*
Autos	1.034*	0.240*	−0.394*
Aero	1.077*	0.154*	−0.809*
Ships	0.956*	0.401*	−0.626*
Guns	1.060*	0.287*	−0.929*
Gold	0.666*	0.801*	−0.399*
Mines	0.994*	0.675*	−0.683*
Coal	0.955*	0.509*	−0.607*
Comps	1.038*	0.389*	−1.225*
Chips	1.273*	0.631*	−1.453*
LabEq	1.380*	0.980*	−1.601*
Paper	1.085*	0.196*	−0.670*
Boxes	0.911*	0.086*	−0.613*
Rtail (n)	1.104*	0.342*	−0.884*
Meals (n)	1.035*	0.315*	−0.830*
Banks (n)	1.177*	0.042*	−0.362*
<i>Panel B: Summary statistics of betas</i>			
Average of mean betas	1.040	0.360	−0.807
Standard deviation of mean betas	0.150	0.396	0.314
t-Test: H_0 : Average of all industries = 0	41.609	5.450	−15.426

Table 4: Equity-factor betas

Table 5

Time-varying currency betas for each industry

The table reports the sample mean of the time-varying currency factor betas estimated from the system of Eqs. (2)–(9). Mean abs is the mean of the absolute betas. STD is standard deviation. The industry portfolios are 31 manufacturing industries and five nontraded goods industries. Nontraded goods industries, using the definitions in Bodnar and Gentry (1993), have (n) to the right of their name. The currency factors are represented by percentage changes in the real US Treasury trade-weighted exchange rate index made up of the currencies of 16 developed countries that are the main trading partners of the US (MAJOR) and the real index of the currencies of the other important trading partners (OITP) from the emerging economies. The indices are in foreign currency per US dollar. Significance at the 1% level is represented by *.

Industry	OITP					MAJOR				
	Mean abs	Mean	STD	Minimum	Maximum	Mean abs	Mean	STD	Minimum	Maximum
Agri	0.375*	−0.343*	0.259	−1.222	0.917	0.354*	−0.354*	0.038	−0.438	−0.238
Food	0.570*	0.437*	0.552	−2.030	2.411	0.218*	−0.212*	0.108	−0.506	0.200
Soda	0.392*	−0.149*	0.517	−1.942	1.481	0.160*	−0.096*	0.167	−0.500	0.240
Beer	0.335*	0.244*	0.368	−1.260	2.006	0.158*	−0.132*	0.123	−0.333	0.166
Smoke	0.574*	0.337*	0.676	−2.251	2.661	0.116*	0.105*	0.093	−0.152	0.363
Toys	0.597*	−0.204*	0.785	−4.084	2.114	0.239*	−0.164*	0.239	−0.559	0.677
Fun (n)	0.169*	0.073*	0.229	−1.240	0.914	0.119*	0.119*	0.060	0.003	0.278
Books	0.319*	0.151*	0.402	−2.207	1.285	0.202*	0.202*	0.118	−0.026	0.459
Hshld	0.339*	0.126*	0.445	−2.128	1.381	0.103*	−0.023*	0.122	−0.236	0.316
Clths	0.541*	−0.077	0.766	−5.160	2.605	0.113*	0.053*	0.128	−0.299	0.375
MedEq	0.331*	0.267*	0.306	−0.969	0.987	0.173*	−0.172*	0.065	−0.332	0.034
Drugs	0.406*	0.336*	0.350	−0.838	1.805	0.333*	−0.331*	0.172	−0.702	0.126
Chems	0.456*	0.076*	0.641	−3.856	2.542	0.371*	−0.371*	0.065	−0.492	−0.076
Rubbr	0.621*	−0.563*	0.523	−2.669	2.599	0.253*	−0.212*	0.206	−0.633	0.342
Txltls	0.600*	−0.135*	0.824	−5.285	3.241	0.096*	−0.043*	0.128	−0.491	0.298
BldMt	0.391*	−0.246*	0.468	−1.929	2.492	0.184*	−0.087*	0.194	−0.463	0.420
Cnst (n)	0.511*	−0.286*	0.576	−3.270	1.412	0.144*	−0.033*	0.170	−0.418	0.330
Steel	0.557*	−0.473*	0.466	−2.803	1.082	0.301*	−0.300*	0.123	−0.472	0.029
FabPr	0.473*	−0.409*	0.405	−1.822	1.410	0.022*	−0.014*	0.025	−0.095	0.158
Mach	0.589*	−0.566*	0.355	−1.999	0.565	0.462*	−0.462*	0.081	−0.592	−0.195
ElcEq	0.892*	−0.717*	0.883	−3.982	3.872	0.282*	−0.101*	0.331	−0.908	0.949
Autos	0.477*	0.303*	0.539	−2.644	2.058	0.067*	−0.037*	0.070	−0.186	0.167
Aero	0.369*	0.273*	0.357	−1.660	1.470	0.133*	−0.030*	0.160	−0.608	0.341
Ships	0.622*	0.170*	0.819	−4.308	2.107	0.406*	0.404*	0.117	−0.114	0.765
Guns	0.829*	0.076	1.116	−4.601	5.592	0.342*	−0.266*	0.319	−1.228	1.385
Gold	0.894*	−0.875*	0.482	−2.124	0.351	1.107*	−1.107*	0.167	−1.515	−0.862
Mines	0.324*	−0.270*	0.274	−1.077	1.137	0.705*	−0.705*	0.199	−1.108	−0.250
Coal	0.426*	−0.366*	0.350	−1.967	0.647	0.116*	0.114*	0.046	−0.138	0.207
Comps	0.460*	−0.159*	0.563	−2.753	1.462	0.269*	−0.212*	0.211	−0.548	0.640
Chips	0.681*	−0.360*	0.789	−3.915	2.312	0.205*	−0.101*	0.238	−0.535	0.686
LabEq	1.152*	−0.974*	0.960	−4.755	1.592	0.100*	0.078*	0.099	−0.230	0.459
Paper	0.205*	0.085*	0.278	−1.285	1.263	0.196*	−0.196*	0.060	−0.277	−0.029
Boxes	0.360*	−0.027	0.505	−3.145	2.252	0.272*	−0.270*	0.116	−0.448	0.111
Rtail (n)	0.379*	0.254*	0.415	−2.245	1.372	0.185*	0.185*	0.072	0.060	0.334
Meals (n)	0.256*	−0.136*	0.317	−1.120	1.384	0.199*	−0.080*	0.222	−0.414	0.471
Banks (n)	0.238*	0.213*	0.177	−0.469	0.985	0.135*	0.135*	0.039	0.059	0.222
Average	0.492	−0.109				0.246	−0.131			
Standard deviation	0.209	0.358				0.197	0.269			
t-Statistic H_0 : avg = 0	14.141	−1.823				7.472	−2.927			

Table 5: Time-varying currency betas

premium is the sum of the estimated expected risk

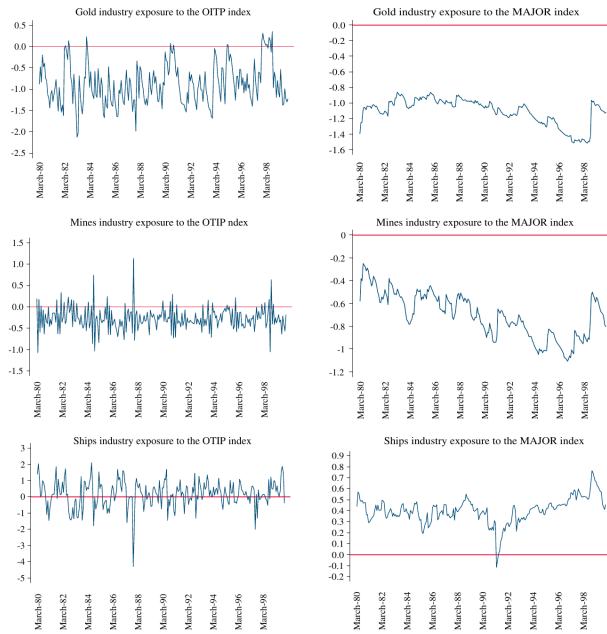


Fig. 1. Time-varying currency betas for a subsample of industries. MAJOR is a currency risk factor, represented by percentage changes in the real US Treasury trade-weighted exchange rate index of the currencies of 16 developed countries that are the main trading partners of the United States. OTIP is a similar risk factor made up of the currencies of 19 other important trading partners from the emerging economies. Both indices are expressed as foreign currency per US dollar.

Figure 1: Time-varying currency betas, selected industries, part A

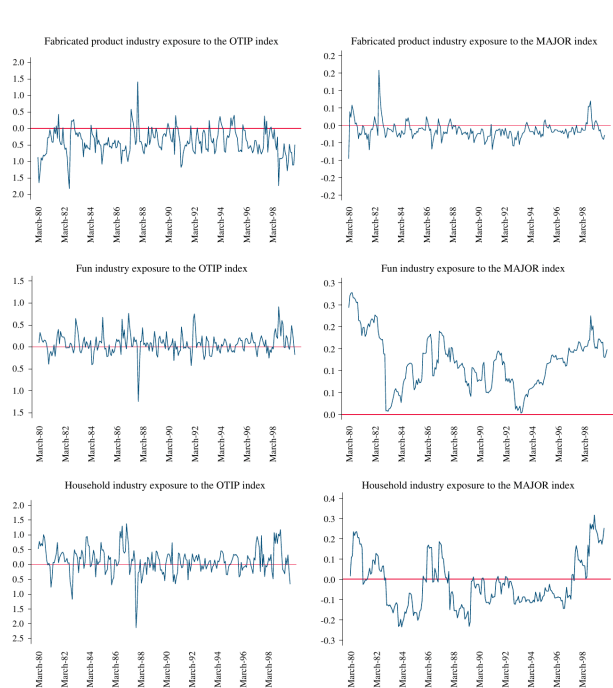


Figure 2: Time-varying currency betas, selected industries, part B

5.4 Tables 6–7 and Figure 2: is currency risk priced and economically important?

Read:

- Table 6 tests whether conditionally expected factor returns are zero. The currency factors are rejected as unpriced in essentially all industries.
- Table 7 converts betas and expected factor returns into risk premia.
- The average absolute currency premium is 2.47 percentage points per year across industries.
- The average currency premium accounts for 11.7% of total absolute risk premium.
- Figure 2 shows that currency risk accounts for more than 10% of total risk premium in many industries and reaches about one-third in the gold industry.

Economic message: Currency risk is not a small add-on. It is a meaningful part of industry cost of equity, even after controlling for market, size, and value factors.

Table 6
Hypothesis tests of the pricing of currency and other risks

The table reports p-values of tests of the null hypothesis that the conditionally expected returns of the risk factors are zero [the coefficients in Eqs. (3)–(7) are zero] and so the risk factors are not priced. All hypotheses are based on Wald tests made robust to non-normality of the residuals. The instruments for the equity factors are the one-period lagged DEFAULT, TERM, and FED variables, as well as the lagged dependent variable (the risk factor). For the currencies, the instruments are FED, EXPRATIO, IMPRATIO, and the lagged dependent variable. All variables are defined in Tables 1 and 3.

Industry	H ₀ : VWM expected return = 0	H ₀ : SMB expected return = 0	H ₀ : HML expected return = 0	H ₀ : Equity factors expected returns = 0	H ₀ : OTP expected return = 0	H ₀ : MAJOR expected return = 0	H ₀ : Currency factors expected returns = 0	H ₀ : All risk factors expected returns = 0
Agri	0.028	0.000	0.000	0.000	0.031	0.000	0.000	0.000
Food	0.025	0.000	0.009	0.000	0.073	0.000	0.000	0.000
Soda	0.017	0.000	0.008	0.000	0.007	0.000	0.000	0.000
Beer	0.000	0.000	0.000	0.000	0.022	0.000	0.000	0.000
Smoke	0.011	0.000	0.000	0.000	0.057	0.000	0.000	0.000
Toys	0.017	0.000	0.000	0.000	0.026	0.000	0.000	0.000
Fun (n)	0.219	0.000	0.045	0.000	0.038	0.000	0.000	0.000
Books	0.006	0.000	0.002	0.000	0.083	0.000	0.000	0.000
Hshld	0.013	0.000	0.000	0.000	0.102	0.000	0.000	0.000
Clths	0.006	0.000	0.000	0.000	0.188	0.000	0.000	0.000
MedEq	0.007	0.000	0.000	0.000	0.057	0.000	0.000	0.000
Drugs	0.020	0.000	0.001	0.000	0.000	0.000	0.000	0.000
Chemis	0.131	0.000	0.010	0.000	0.012	0.000	0.000	0.000
Rubbr	0.014	0.000	0.047	0.000	0.014	0.000	0.000	0.000
Textls	0.008	0.000	0.054	0.000	0.019	0.000	0.000	0.000
BldMnt	0.000	0.000	0.020	0.000	0.354	0.000	0.000	0.000
Cnst (n)	0.014	0.000	0.002	0.000	0.060	0.000	0.000	0.000
Steel	0.067	0.000	0.002	0.000	0.011	0.000	0.000	0.000
FabPr	0.025	0.000	0.000	0.000	0.009	0.000	0.000	0.000
Mach	0.029	0.000	0.024	0.000	0.014	0.000	0.000	0.000
ElcEq	0.086	0.000	0.289	0.000	0.017	0.000	0.000	0.000
Autos	0.002	0.000	0.000	0.000	0.066	0.000	0.000	0.000
Aero	0.006	0.000	0.008	0.000	0.001	0.000	0.000	0.000
Ships	0.008	0.000	0.013	0.000	0.028	0.000	0.000	0.000
Guns	0.001	0.000	0.000	0.000	0.015	0.000	0.000	0.000
Gold	0.000	0.000	0.000	0.000	0.077	0.000	0.000	0.000
Mines	0.010	0.000	0.001	0.000	0.017	0.000	0.000	0.000
Coal	0.029	0.000	0.000	0.000	0.009	0.000	0.000	0.000
Comps	0.006	0.000	0.000	0.000	0.010	0.000	0.000	0.000
Chips	0.229	0.000	0.076	0.000	0.007	0.000	0.000	0.000
LabEq	0.041	0.000	0.045	0.000	0.028	0.000	0.000	0.000
Paper	0.078	0.000	0.048	0.000	0.014	0.000	0.000	0.000
Boxes	0.044	0.000	0.050	0.000	0.031	0.000	0.000	0.000
Rtail (n)	0.023	0.000	0.004	0.000	0.137	0.000	0.000	0.000
Meals (n)	0.012	0.000	0.161	0.000	0.068	0.000	0.000	0.000
Banks (n)	0.004	0.000	0.031	0.000	0.038	0.000	0.000	0.000

Table 7
Estimates of time-varying risk premiums

Panel A of the table reports the sample mean risk premium associated with each risk factor, the sum of the individual currency premiums, the total risk premium, and the percentage of the total risk premium attributed to the currency risk premiums, respectively. The risk premium is the product of the respective beta and corresponding expected return on the factor for each period. The total risk premium is the sum of the three equity and two currency risk premiums, in each period. For comparison across premiums, all premiums are reported as the mean of the absolute period-by-period premium. The risk factors are returns on the Fama and French (1993) factors: excess returns on the Center for Research in Security Prices value-weighted market index (VWM), returns on small minus big firms (SMB), and returns on high minus low book-to-market value firms (HML). The currency factors are represented by percentage changes in the real US Treasury trade-weighted exchange rate index made up of the currencies of 16 developed countries that are the main trading partners of the US (MAJOR) and the real index of the currencies of the other important trading partners (OTP) from the emerging economies, expressed as foreign currency per US dollar. Panel B reports summary statistics across all 36 industries. Significance at the 1% level is represented by *.

Industry	Market	SMB	HML	OTP	MAJOR	Total currency	Total premium	Currency as a percent of total
<i>Panel A: Summary of absolute time-varying risk premiums</i>								
Agri	10.309*	4.078*	4.768*	0.693*	1.949*	2.643*	21.798*	12.971
Food	8.917*	1.829*	2.024*	1.047*	1.98*	2.245*	15.015*	16.200
Soda	11.022*	2.194*	3.356*	1.364*	0.889*	2.352*	18.825*	13.161
Beer	9.056*	1.197*	4.024*	0.719*	0.700*	1.508*	15.786*	10.980
Smoke	10.441*	1.288*	2.269*	1.049*	0.646*	1.695*	15.692*	12.336
Toys	14.818*	4.082*	7.784*	1.345*	1.280*	2.625*	29.110*	9.145
Fun (n)	10.679*	2.508*	4.348*	0.365*	0.718*	1.083*	18.618*	6.013
Books	12.671*	1.282*	3.946*	0.631*	1.211*	1.846*	19.745*	9.757
Hshld	13.317*	1.121*	4.759*	0.585*	0.524*	1.110*	20.307*	6.061
Clths	12.900*	4.814*	6.225*	0.987*	0.751*	1.738*	25.678*	7.192
MedEq	11.363*	0.414*	7.028*	0.721*	0.937*	1.658*	20.463*	9.266
Drugs	12.437*	1.716*	4.936*	1.176*	1.721*	2.897*	21.986*	14.676
Chemis	11.874*	1.032*	3.690*	1.165*	2.112*	3.278*	19.874*	17.052
Rubbr	13.637*	5.320*	3.868*	1.558*	1.332*	2.890*	25.715*	12.292
Textls	13.811*	6.438*	3.351*	1.114*	0.662*	1.976*	25.576*	8.105
BldMnt	16.213*	1.940*	3.806*	0.942*	1.021*	1.963*	23.922*	9.343
Cnst (n)	12.449*	5.374*	6.703*	0.991*	0.738*	1.729*	26.254*	6.909
Steel	11.913*	3.865*	4.975*	1.212*	1.642*	2.854*	23.347*	13.778
FabPr	12.112*	5.221*	4.945*	1.174*	0.141*	1.315*	23.593*	5.683
Mach	11.762*	3.668*	5.691*	1.223*	2.565*	3.788*	24.908*	15.681
ElcEq	12.531*	6.318*	4.707*	2.189*	1.733*	3.962*	26.981*	14.733
Autos	12.528*	2.098*	3.352*	0.968*	0.365*	1.332*	19.312*	7.859
Aero	12.601*	1.287*	5.068*	0.913*	0.785*	1.698*	20.654*	8.682
Ships	10.917*	2.790*	4.517*	1.516*	2.267*	3.782*	22.006*	17.538
Guns	7.742*	3.135*	7.050*	2.400*	2.095*	4.496*	32.421*	14.682
Gold	9.005*	5.313*	3.693*	1.869*	5.913*	7.781*	25.793*	32.937
Mines	12.235*	4.887*	5.227*	0.715*	3.979*	4.693*	27.043*	18.543
Coal	9.988*	3.251*	4.508*	0.942*	0.660*	1.602*	19.359*	8.488
Comps	12.801*	2.679*	8.353*	1.033*	1.409*	2.443*	26.275*	10.108
Chips	14.071*	4.037*	5.720*	1.552*	1.043*	2.595*	26.423*	9.808
LabEq	15.663*	6.070*	7.950*	2.265*	0.563*	2.828*	32.510*	8.855
Paper	11.248*	1.374*	4.129*	0.456*	1.046*	1.502*	18.254*	9.112
Boxes	10.166*	1.069*	3.513*	0.895*	1.477*	2.312*	17.060*	14.661
Rtail (n)	13.860*	2.129*	4.984*	0.743*	1.057*	1.800*	22.574*	8.580
Meals (n)	10.834*	2.021*	3.336*	0.680*	1.126*	1.805*	17.996*	11.292
Banks (n)	14.459*	0.390*	1.928*	0.502*	0.678*	1.180*	17.957*	8.946
<i>Panel B: Averages across all industries</i>								
Average	12.288*	3.005*	4.722*	1.108*	1.361*	2.470*	22.484*	11.707
Standard deviation	2.000	1.784	1.588	0.493	1.072	1.307	4.449	5.049
Minimum	8.917	0.390	1.928	0.365	0.141	1.083	15.015	5.683
Maximum	17.742	6.438	8.353	2.400	5.913	7.781	32.510	32.937

Table 6: Hypothesis tests of pricing of currency and other risks

Table 7: Estimates of time-varying risk premia

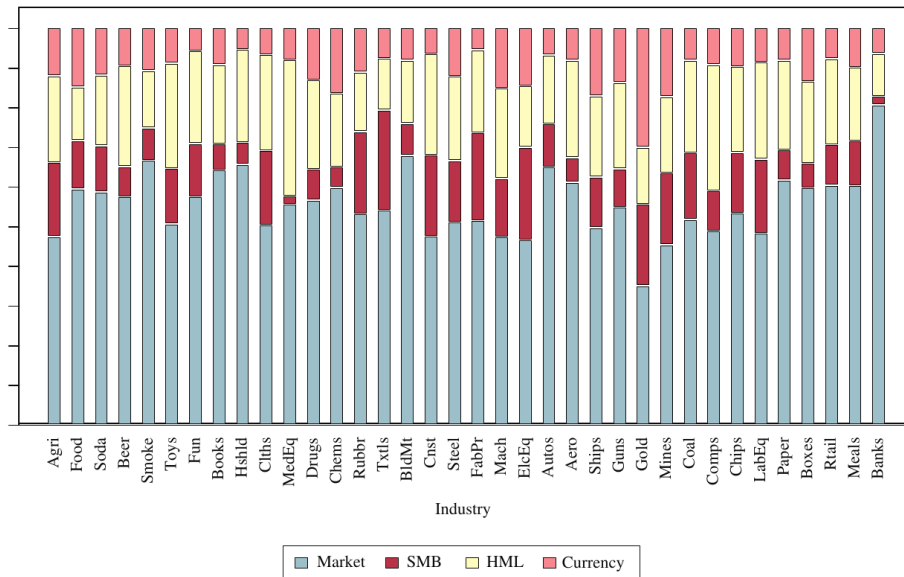


Figure 3: Currency risk premium as a percentage of total risk premium

Currency risk premium as a percentage of total risk premium for different industries. Currency risk premium is the sum of the risk premiums of the risk factor, represented by percentage changes in the real US Treasury trade-weighted exchange rate index of the currencies of 16 developed countries that are the main trading partners of the United States and the OTP risk factor, made up of the currencies of 19 other important trading partners from emerging economies. Both indices are expressed as foreign currency per US dollar. The other risk premiums are related to the excess return on the value-weighted market index (VWM), returns on small minus big firms (SMB), and returns on high minus low book-to-market value firms (HML).

5.5 Figure 3 and Table 8: time variation and subperiod evidence

Read:

- Figure 3 plots currency premia for selected industries and shows large time variation.

- The total currency premium changes sign for some industries and rises sharply in particular periods.
- Table 8 splits the sample into 1980–1989 and 1990–1999.
- Both subperiods show significant currency betas and premia, suggesting that the result is not a single-decade artifact.

Economic message: The currency premium is a time-varying component of cost of equity. Its importance persists across subperiods but changes in magnitude and composition.



currency risk premium for a subsample of industries. MAJOR is the real US Treasury trade-weighted exchange rate index of the 19 countries that are the main trading partners of the United States. OITP is a similar index made up of the currencies of 19 countries from the emerging economies. Both indices are measured as foreign currency per US dollar.

Figure 4: Currency risk premia, selected industries, part A

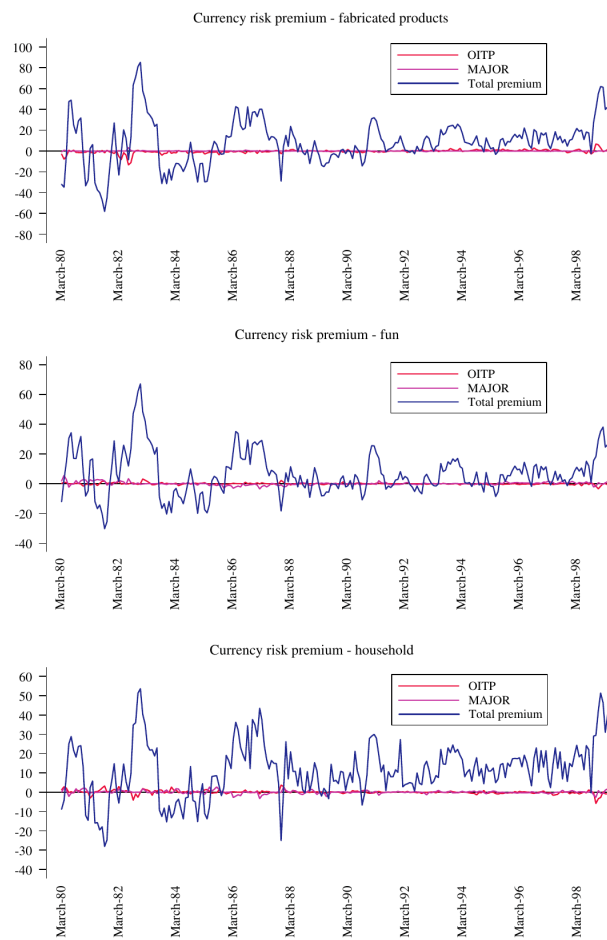


Figure 5: Currency risk premia, selected industries, part B

5.6 Table 9: residual diagnostics and model fit

Read:

- Table 9 checks residual diagnostics for the conditional mean model.
- The tests examine residual means, normality, autocorrelation, squared-residual autocorrelation, predictability, and mean pricing error.
- The mean pricing error is generally not statistically different from zero.

Table 8
Subperiod exchange rate risk
Panel A of this table reports the sample mean of the absolute currency betas and absolute currency risk premiums for each industry in the periods January 1980–December 1989 and January 1990–December 1999. The risk premium is the product of the currency beta and expected change in the currency factor for each period. Panel B is a summary across all 36 industries. Significance at the 1% level is represented by *.

Industry	1980–1989					1990–1999				
	OITP beta	MAJOR beta	Total currency premium	Total premium	Percent of total premium	OITP beta	MAJOR beta	Total currency premium	Total premium	Percent of total premium
Panel A: Mean time-varying exchange rate betas and risk premiums										
Agri	0.378*	0.345*	3.009*	23.719*	14.025	0.371*	0.362*	2.283*	19.910*	11.934
Food	0.025*	0.221*	2.707*	10.048*	19.206	0.505*	0.216*	1.791*	13.999*	13.245
Soda	0.353*	0.187*	2.632*	19.303*	15.880	0.430*	0.134*	1.879*	18.356*	10.488
Beer	0.389*	0.109*	1.540*	18.011*	9.823	0.281*	0.206*	1.478*	13.598*	12.118
Smoke	0.563*	0.101*	1.761*	16.544*	13.613	0.584*	0.130*	1.630*	14.855*	11.060
Toys	0.622*	0.227*	2.943*	32.894*	9.622	0.573*	0.250*	2.312*	2.312*	8.676
Fun (n)	0.177*	0.132*	1.359*	21.057*	7.009	0.162*	0.107*	0.812*	16.219*	5.033
Books	0.381*	0.228*	2.444*	23.368*	11.433	0.258*	0.177*	1.258*	16.182*	8.110
Hshld	0.390*	0.113*	1.373*	20.957*	7.726	0.290*	0.093*	0.851*	19.667*	4.423
Clths	0.568*	0.133*	2.132*	30.748*	8.086	0.515*	0.094*	1.351*	20.692*	6.313
MedEq	0.361*	0.166*	1.925*	20.130*	11.144	0.303*	0.180*	1.396*	20.791*	7.419
Drugs	0.440*	0.304*	3.281*	21.794*	16.824	0.372*	0.361*	2.520*	22.176*	12.564
Chem	0.513*	0.354*	3.767*	21.698*	18.537	0.400*	0.388*	2.796*	18.081*	15.592
Rubber	0.633*	0.196*	2.945*	27.811*	12.244	0.608*	0.338*	2.836*	23.654*	12.339
Txits	0.621*	0.129*	2.519*	29.748*	9.679	0.580*	0.063*	1.441*	21.473*	6.557
BlldMt	0.454*	0.155*	2.275*	27.222*	10.009	0.330*	0.212*	1.656*	20.677*	8.689
Cost (n)	0.552*	0.125*	2.048*	30.298*	7.449	0.471*	0.162*	1.415*	22.278*	6.377
Steel	0.604*	0.251*	3.212*	26.482*	13.859	0.511*	0.350*	2.502*	20.263*	13.697
FabPr	0.487*	0.022*	1.516*	26.236*	6.236	0.461*	0.022*	1.117*	20.994*	5.139
Mach	0.630*	0.409*	4.240*	27.644*	16.163	0.549*	0.513*	3.343*	22.219*	15.206
ElcEq	0.942*	0.251*	4.283*	29.002*	15.311	0.842*	0.313*	3.946*	24.994*	14.164
Autos	0.501*	0.057*	1.415*	23.182*	7.232	0.452*	0.077*	1.251*	15.505*	8.475
Aero	0.391*	0.129*	1.929*	22.332*	9.591	0.349*	0.136*	1.471*	19.004*	7.788
Ships	0.731*	0.397*	4.665*	27.128*	17.913	0.516*	0.414*	2.914*	16.971*	17.170
Guns	0.873*	0.373*	5.527*	33.695*	17.949	0.787*	0.312*	3.482*	31.711*	11.470
Gold	1.000*	1.001*	8.926*	32.559*	31.172	0.790*	1.211*	6.656*	19.139*	34.673
Mines	0.341*	0.554*	4.738*	26.799*	18.561	0.306*	0.853*	4.650*	25.316*	18.325
Coal	0.449*	0.141*	2.079*	22.178*	9.955	0.404*	0.091*	1.134*	16.587*	7.045
Comps	0.408*	0.257*	2.847*	29.703*	10.771	0.423*	0.281*	2.046*	22.904*	9.455
Chips	0.699*	0.147*	2.569*	27.141*	9.972	0.664*	0.262*	2.621*	25.716*	9.144
LabEq	1.201*	0.091*	3.268*	14.997*	9.347	1.103*	0.109*	2.394*	28.074*	8.452
Paper	0.227*	0.171*	1.609*	20.121*	9.214	0.182*	0.221*	1.396*	16.418*	9.011
Boxes	0.211*	0.259*	2.467*	19.794*	14.844	0.348*	0.285*	2.141*	14.982*	14.482
Retail (n)	0.411*	0.190*	2.151*	25.389*	9.738	0.348*	0.180*	1.454*	20.598*	7.440
Meats (n)	0.267*	0.170*	1.979*	20.591*	11.086	0.246*	0.227*	1.634*	15.445*	11.495
Banks (n)	0.250*	0.117*	1.278*	18.600*	10.199	0.227*	0.152*	1.084*	17.264*	7.714
Panel B: Summary results across all industries										
Mean	0.525*	0.227*	2.816*	24.343*	12.537	0.459*	0.263*	2.129*	19.402*	10.891*
Standard deviation	0.221	0.174	1.483	5.967	4.917	0.200	0.224	1.166	4.973	5.381
Minimum	0.177	0.022	1.278	14.997	6.236	0.162	0.022	0.812	2.312	4.423
Maximum	1.201	1.001	8.926	33.695	31.172	1.103	1.211	6.656	31.711	34.673

Table 9
Industry residual diagnostics
The table reports diagnostic tests on the residuals of the conditional mean model of the industry portfolio, u_{it} from Eq. (2). The tests in the first four columns are conducted on the residuals standardized by the conditional standard deviation of the industry returns obtained from Eq. (9). JB is the Jarque-Bera test of the null hypothesis that the standardized residuals are normally distributed. $Q(12)$ is the Q-statistic of the test of the null hypothesis that the standardized residuals are not autocorrelated up to the 12th lag, and $Q^2(12)$ tests for autocorrelation in the squared residuals (i.e., a test for remaining heteroskedasticity in the residuals). The test in the fifth column tests the null hypothesis that the raw residuals from the conditional mean equation of the industry returns are not predictable, using the lagged instruments DEFAULT, TERM, FED, and VWM defined in Tables 1 and 3. The values reported under the first five columns are p-values. The sixth column reports the mean pricing error defined as the realized excess returns–estimated expected excess returns. * And ** represent significance at the 5% and 10% levels, respectively.

Industry	Industry portfolio residuals standardized by conditional standard deviation				Unstandardized residuals: H_0 : Residuals not predictable	Realized excess returns–estimated expected excess returns; mean pricing error
	H_0 : Mean = 0	H_0 : Not normal (JB = 0)	H_0 : Not autocorrelated ($Q(12)$)	H_0 : Squared residuals not autocorrelated ($Q^2(12)$)		
Agri	0.580	0.000**	0.896	0.903	0.590	0.227
Food	0.327	0.000**	0.028**	0.614	0.160	0.271
Soda	0.085	0.000**	0.499	0.881	0.088***	0.485
Beer	0.117	0.002**	0.038**	0.001**	0.371	0.512
Smoke	0.216	0.007**	0.360	0.576	0.268	0.407
Toys	0.178	0.000**	0.255	0.980	0.895	0.669
Fun (n)	0.390	0.000**	0.139	0.760	0.758	0.398
Books	0.440	0.000**	0.395	0.979	0.886	0.233
Hshld	0.589	0.000**	0.376	0.952	0.542	0.116
Clths	0.669	0.000**	0.315	0.983	0.506	-0.180
MedEq	0.307	0.012**	0.031**	0.765	0.441	0.332
Drugs	0.593	0.005**	0.478	0.869	0.609	0.106
Chem	0.668	0.000**	0.318	0.946	0.783	0.075
Rubber	0.722	0.000**	0.522	0.992	0.999	-0.105
Txits	0.213	0.000**	0.117	0.999	0.508	-0.392
BlldMt	0.318	0.000**	0.316	0.847	0.856	-0.258
Cost (n)	0.602	0.000**	0.157	0.767	0.774	0.028
Steel	0.799	0.000**	0.056	0.598	0.418	0.006
FabPr	0.259	0.000**	0.801	0.960	0.698	-0.409
Mach	0.642	0.000**	0.286	0.869	0.957	0.195
ElcEq	0.663	0.000**	0.460	0.947	0.296	0.272
Autos	0.789	0.000**	0.133	0.622	0.096	0.096
Aero	0.713	0.000**	0.041**	0.852	0.931	0.223
Ships	0.711	0.000**	0.531	0.944	0.125	0.243
Guns	0.124	0.000**	0.023**	0.273	0.109***	-0.704
Gold	0.726	0.000**	0.472	0.753	0.665	-0.012
Mines	0.411	0.000**	0.812	0.417	0.932	-0.235
Coal	0.444	0.000**	0.708	0.952	0.586	-0.269
Comps	0.410	0.013**	0.304	0.402	0.754	0.305
Chips	0.322	0.000**	0.706	0.759	0.482	0.450
LabEq	0.474	0.029**	0.027**	0.447	0.918	0.379
Paper	0.613	0.000**	0.897***	0.001**	0.808	0.006
Boxes	0.708	0.000**	0.155	0.805	0.572	0.102
Rtail (n)	0.450	0.000**	0.136	0.996	0.983	0.229
Meats (n)	0.915	0.000**	0.239	0.988	0.868	0.062
Banks (n)	0.957	0.000**	0.343	0.996	0.828	-0.073

Economic message: The diagnostics support the conditional asset-pricing specification. The model does not appear to be generating large unexplained pricing errors.

5.7 Table 10: cross-sectional determinants of currency risk premia

Read:

- Table 10 regresses industry currency premia on industry characteristics.
- Foreign income over total sales is positive in many specifications, indicating that internationally exposed industries have larger currency premia.
- The acid ratio is often negative, suggesting that liquidity can reduce currency-risk costs.
- Leverage and industry concentration also help explain currency premia.
- Adjusted R^2 values rise as more characteristics are added, reaching roughly 27% for OITP and 17% for MAJOR in the full specifications.

Economic message: The currency premium is not random noise. It is connected to observable industry characteristics related to international exposure, market structure, financing constraints, and balance-sheet risk.

5.8 Table 11: macro determinants of time variation

Read:

- Table 11 estimates industry-by-industry regressions of currency premia on macro variables.
- The Volcker period, foreign trade ratios, growth opportunities, and crisis dummies frequently enter significantly.
- The explanatory power varies widely across industries; some adjusted R^2 values are low, while others exceed 70%.
- The table is presented in two parts because the original table spans two pages.

Table 10
Explaining the cross-sectional variation in currency risk premiums

The table reports ordinary least squares tests of the explanatory power of industry characteristics for the cross-sectional variation in industry currency risk premium obtained from the currency factors. The currency factors are represented by percentage changes in the real US Treasury trade-weighted exchange rate index made up of the currencies of 16 developed countries that are the main trading partners of the US (MAJOR) and the real index of the currencies of the other important trading partners (OITP) from the emerging economies, expressed as foreign currency per US dollar. In Panel A the dependent variable is the annual average of the estimated monthly time-varying OITP currency betas. In Panel B the dependent variable is the annual average of the estimated monthly time-varying MAJOR currency betas. The continuous independent variables are computed annually from firm-level variables within each industry. They are book-to-market ratio, leverage, acid ratio, Herfindahl index of net sales, foreign income as a fraction of net sales, and log of market equity. We also include dummy variables defined as one for each nontraded industry and zero otherwise. Nontraded industries are as defined in Bodnar and Gentry (1993). Significance is based on White standard errors. The sample period is January 1980–December 1999, for a total of 720 industry years (20 years times 36 industries). However, foreign income is not available in some years, resulting in a total of 570 observations. Significance at the 1%, 5%, and 10% levels are represented by *, **, and ***, respectively.

Panel A: Risk premium arising from developing countries' currencies

Independent variable	OITP						
	Model (1)	Model (2)	Model (3)	Model (4)	Model (5)	Model (6)	Model (7)
Constant	0.285*	-0.155	-0.180	-1.024*	-2.501*	-1.195*	-1.876*
Acid ratio	-0.318*	-0.331*	-0.326*	-0.437*	-0.430*	-0.430*	-0.334*
Foreign income/total sales	2.723*	1.811*	1.815*	1.620*	0.940	1.048*	1.009***
Herfindahl index of industry sales	0.151	-0.060	-0.061	0.620*	0.534*	0.385*	0.422*
Market-to-book ratio		0.213*			0.157*	0.170*	0.161**
Log(market equity)				0.125*	0.124*	0.107*	0.124*
Leverage			0.062		1.940*		1.299*
Construction	0.135	0.228**	0.217***	0.394*	-0.109	0.433*	0.237***
Fun	0.325*	0.277*	0.270*	0.328*	0.019	0.290*	0.138**
Retail	0.266*	0.221*	0.223*	-0.004	0.197*	-0.002	0.012
Meals	-0.106***	-0.111***	-0.113***	-0.126*	-0.064	-0.127**	-0.161*
Banks	0.856*	1.045*	1.042*	0.854*	0.432*	1.006*	0.921*
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	570	570	570	570	570	570	570
Adjusted R ²	10.43	18.40	18.25	21.09	25.04	25.96	27.12

Panel B: Risk premium arising from industrialized countries' currencies

Independent variable	MAJOR						
	Model (1)	Model (2)	Model (3)	Model (4)	Model (5)	Model (6)	Model (7)
Constant	0.068	-0.039	-0.255	0.335*	0.033	0.402*	0.098
Acid ratio	-0.101	-0.104	-0.032	-0.068	-0.071	-0.071	-0.035
Foreign income/total sales	1.414***	1.192	1.456***	1.467***	1.446***	1.694**	1.453***
Herfindahl index of industry sales	0.134***	0.082	0.111	-0.078	-0.053	0.014	-0.065
Market-to-book ratio		0.052*		0.067*	0.063*		0.064*
Log(market equity)				-0.039*	-0.033*	-0.032*	-0.033*
Leverage			0.802*	0.518*	0.518*	0.451***	
Construction	0.226*	0.249*	0.086	0.175**	0.071**	0.160***	0.107
Fun	0.348*	0.336*	0.252*	0.331*	0.286*	0.347*	0.279*
Retail	0.538*	0.527*	0.369***	0.408*	0.432*	0.407*	0.413*
Meals	0.000	-0.001	-0.019	0.005	0.003	0.005	-0.007
Banks	0.486*	0.532*	0.439*	0.546*	0.466*	0.486*	0.517*
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	570	570	570	570	570	570	570
Adjusted R ²	13.71	14.57	14.74	16.52	16.79	15.03	16.69

Table 11
Explaining the time variation in currency risk premiums

The table reports ordinary least squares tests of the explanatory power of macro variables for the time-series variation in industry currency risk premium that are obtained from the currency factors. The currency factors are represented by percentage changes in the real US Treasury trade-weighted exchange rate index made up of the currencies of 16 developed countries that are the main trading partners of the US (MAJOR) and the real index of the currencies of the other important trading partners (OITP) from the emerging economies, expressed as foreign currency per US dollar. The dependent variables are monthly OITP and MAJOR currency betas. The independent variables are a dummy variable defined as one for the months identified by the National Bureau of Economic Research (NBER) as the Mexican currency crisis (MEXCRASH), the Mexican currency crisis (MEXCRASH) represented as one during the period December 1994–June 1995 and zero otherwise; the Asian currency crisis (ASACRASH) represented as one during the period June 1997–December 1997 and zero otherwise; the Volcker experiment in tight monetary policy represented as one for the period October 1979–October 1982 and zero otherwise; US monthly aggregate exports as a percentage of quarterly gross domestic product (GDP) (EXPRATIO); US monthly aggregate imports as a percentage of quarterly GDP (IMPRATIO); and aggregate growth opportunities (GROWOPP) represented by the price-to-earnings ratio of the Standard & Poor's 500 index. The continuous variables are standardized (the mean is subtracted and the remainder divided by the sample standard deviation) and enter the regressions contemporaneously and with two lags. The reported coefficients are the sum of the three coefficients and the associated p-value tests the hypothesis that the sum is equal to zero. All p-values are based on Newey and West heteroskedasticity- and autocorrelation-consistent (with one lag) standard errors. *, **, and *** represent significance at the 1%, 5%, and 10% level, respectively. The data are monthly from December 1979 to December 1999, but the asset pricing model uses the first three observations in the estimation, hence, the betas are from March 1980 to December 1999 (237 observations).

Industry	Constant	BUSCYCLE	MEXCRASH	ASACRASH	Volker experiment	EXPRATIO	IMPRATIO	GROWOPP	χ^2 (14)	Adjusted R ²
Panel A: Time variation in OITP currency risk premium										
Agri	-0.380*	0.082	0.209*	-0.088	0.144***	0.034	0.012	0.084*	0.001	9.80
Food	0.423*	-0.158	-0.409*	0.323**	0.153	-0.051	0.081	-0.006	0.000	5.27
Soda	-0.118*	0.198	0.000	0.092	-0.273***	-0.066	-0.087	-0.126**	0.047	8.97
Beer	0.223*	-0.184**	-0.340*	-0.112	0.252*	0.025	-0.020	0.020	0.000	10.03
Smoke	0.298*	-0.078	-0.134	-0.054	0.220	0.002	0.073	0.205*	0.080	6.51
Toys	-0.179*	0.174	-0.134	-0.777**	-0.401***	0.050	0.012	-0.151	0.169	7.32
Fun (n)	0.068*	0.038	-0.051	-0.135**	-0.018	0.015	-0.007	0.062**	0.036	5.40
Books	0.136*	-0.139	0.090	-0.238**	0.104	0.007	-0.009	0.055	0.004	5.62
Hshld	0.109*	-0.086	-0.175*	0.078	0.030	-0.018	0.005	-0.002	0.010	5.95
Clths	-0.080	-0.376**	-0.226	-0.592*	0.317	0.081	0.055	0.043	0.008	8.78
MedEq	0.266*	-0.013	0.382*	-0.045	0.028	-0.087	0.019*	-0.077**	0.010	5.90
Drugs	0.368*	-0.177	-0.067	-0.115	0.067	-0.019	-0.029	0.094**	0.000	9.63
Chem	0.036	0.046	0.130	-0.302	0.177	0.000	0.156**	0.108	0.025	6.27
Rubbr	-0.339*	0.038	0.134	-0.294	-0.143	0.020	-0.077	-0.153**	0.019	10.19
Txtils	-0.209*	0.005	0.260	-0.770*	0.441	-0.003	0.186**	0.207**	0.007	8.13
BldMnt	-0.248*	-0.003	0.247*	0.177	0.062	-0.011	-0.042	-0.039	0.096	1.76
Cnst(n)	0.135*	-0.006	0.244	-0.189	0.189	0.079	-0.015	0.165*	0.190	3.62
Steel	-0.540*	0.254*	0.385**	-0.177	0.150	-0.003	0.114**	0.121**	0.008	9.53
FabPr	-0.396*	0.063	0.441*	0.297*	-0.181	0.020	-0.079**	-0.087*	0.000	7.85
Mach	-0.103*	0.221***	0.372*	-0.078	0.015	0.010	0.054	0.079**	0.006	10.66
ElcEq	-0.748*	0.270	0.518*	1.068**	0.019	0.033	-0.039*	0.028	0.003	9.48
Autos	0.321*	-0.175	-0.165	-0.435**	-0.032	0.006	0.056	0.089	0.023	4.96
Aero	0.284*	-0.071	-0.295*	-0.142	-0.033	0.002	0.062***	-0.013	0.010	2.06
Ships	0.168*	0.229	-0.457**	-0.804*	-0.166	-0.018	0.170	-0.046	0.229	2.54
Guns	0.082	0.046	0.157	-0.221	0.041	-0.008	-0.009	-0.100	0.504	-2.08
Gold	-0.914*	0.487*	0.437**	0.155	-0.247***	0.041*	-0.007	0.017	0.000	13.89
Mines	-0.290*	0.046	0.259*	0.032	0.067	0.009	-0.009	0.021	0.000	1.06
Coal	-0.417*	0.368*	0.090	-0.221	0.043	-0.004	0.073**	0.084**	0.001	12.52
Comp	-0.185*	0.050	0.054	-0.474	0.034	0.026	0.112	-0.027	0.006	5.80
Chips	-0.375*	0.204	0.161	-0.697	-0.195	0.084	0.068	-0.053	0.269	6.42
LabEq	-1.052	0.482	0.753*	-0.676	-0.054	0.050	0.129	0.100	0.002	7.34
Paper	0.072*	-0.053	0.397*	-0.204**	-0.122	-0.008	0.061**	0.081**	0.115	5.80
Roses	-0.059	0.134	-0.082	0.077	0.000	0.009	0.104***	0.093	0.004	6.81
Rtail (n)	0.263*	-0.203***	-0.275**	-0.385*	0.067	-0.026	0.054	-0.006	0.003	10.10
Meals (n)	0.142*	0.098	0.248*	-0.188	-0.027	-0.021	-0.012	0.000	0.762	7.62
Banks (n)	0.240*	-0.065	-0.174*	0.070	-0.037	-0.023	-0.001	-0.054**	0.002	12.13
Panel B: Time variation in MAJOR currency risk premium										
Agri	-0.364*	-0.017	0.012*	-0.042*	0.095*	-0.010*	0.011*	0.040*	0.000	68.41
Food	-0.234*	-0.044	0.000	-0.045**	0.189*	0.023***	0.019***	0.041*	0.000	32.61
Soda	-0.130*	0.100*	0.109*	-0.013	0.121*	0.093*	-0.068*	0.142*	0.000	36.09
Beer	-0.177*	-0.007	0.043**	-0.051	0.346*	-0.050*	0.050*	0.093*	0.000	62.74
Smoke	0.089*	0.014	0.085*	-0.084*	0.114*	0.051*	-0.018*	0.096*	0.000	50.15
Toys	-0.231*	-0.082**	0.190*	0.005	0.501*	0.014	0.139*	0.181*	0.000	71.00
Fun (n)	0.086*	-0.014	0.230	0.010	0.161*	-0.014*	0.034*	0.020*	0.000	94.79
Books	0.155*	-0.084*	0.055*	0.073*	0.377*	0.034*	0.054*	0.074*	0.000	70.83
Hshld	-0.068*	-0.017	0.082*	0.038	0.296*	-0.009	0.048*	0.111*	0.000	65.83
Clths	0.013	-0.098	0.057**	0.007	0.332*	-0.064*	0.098*	0.014	0.000	52.16
MedEq	-0.193*	0.003	0.050*	-0.028*	0.143*	-0.029*	0.040*	0.037*	0.000	43.77
Drugs	-0.383*	-0.011	0.062***	-0.009	0.367*	-0.007	-0.026**	0.149*	0.000	46.59
Chem	-0.389*	-0.042**	0.032***	0.001	0.137*	-0.019*	0.014***	0.003	0.000	31.90

Economic message: The time variation in currency premia is related to macroeconomic forces, monetary policy, trade exposure, and currency-crisis episodes. This strengthens the interpretation that the estimated premia reflect economic exchange-rate risk rather than mechanical model noise.

Table 11 (continued)

Industry	Constant	BUSCYCLE	MEXCRASH	ASACRASH	Volker experiment	EXPRATIO	IMPRATIO	GROWOPP	χ^2 (14)	Adjusted R ²
Rubbr	-0.298*	-0.150*	0.051	0.176*	0.702*	-0.127*	0.104*	0.069*	0.000	74.14
Txtils	-0.001	0.099*	-0.064*	-0.014	-0.357*	0.042*	-0.046*	-0.030**	0.000	48.15
BldMnt	-0.160*	-0.140*	0.081**	0.076*	0.616*	-0.078*	0.085*	0.138*	0.000	68.78
Cnst(n)	-0.101*	-0.125*	0.111*	0.086*	0.543*	-0.089*	0.074*	0.130*	0.000	71.92
Steel	-0.342*	-0.059*	0.055*	-0.045***	0.348*	-0.034*	0.024**	0.084*	0.000	55.82
FabPr	-0.018*	0.000	0.010	-0.001	0.031*	-0.006	0.007**	0.004**	0.019	8.45
Mach	-0.491*	-0.004*	-0.002	0.004	0.253*	-0.031*	0.015**	0.020*	0.000	70.47
ElcEq	-0.203*	-0.197**	0.173**	0.414*	0.729*	-0.107*	0.194*	0.120*	0.000	45.49
Autos	-0.058*	-0.033*	0.027**	-0.003	0.168*	-0.007	0.032*	0.063*	0.000	63.12
Aero	-0.006	0.013	-0.057**	0.067**	-0.157*	-0.101*	-0.015	-0.094*	0.000	58.59
Ships	0.390*	-0.030	-0.024	0.055**	0.106*	-0.027	0.088*	0.011	0.000	33.49
Guns	-0.313*	0.038	-0.054	-0.037	0.224**	0.034	0.071	0.120*	0.000	18.15
Gold	-1.110*	-0.036	0.026	-0.174*	0.113*	-0.100*	-0.035**	0.026	0.000	62.58
Mines	-0.762*	-0.161*	0.028	0.025	0.513*	-0.102*	-0.021	0.082*	0.000	68.31
Coal	0.104	-0.012	0.039*	0.048*	0.083*	-0.036*	0.013*	0.016*	0.000	50.52
Comp	-0.261*	-0.116*	0.128**	-0.023	0.396*	0.021***	0.082*	0.221*	0.000	73.18
Chips	-0.160*	-0.084**	0.087*	-0.151**	0.555*	-0.066*	0.189*	0.155*	0.000	67.50
LabEq	0.062*	-0.053	0.091**	-0.071*	0.147*	-0.032*	0.068*	0.040*	0.000	39.38
Paper	-0.217*	-0.006	0.026*	-0.030*	0.158*	-0.032*	0.024*	0.047*	0.000	61.94
Roses	-0.305*	-0.053*	0.067*	-0.020**	0.292*	-0.054*	0.085*	0.095*	0.000	65.51
Rtail (n)	0.157*	-0.013	0.021**	0.009	0.202*	-0.011	0.015*	0.015*	0.000	75.75
Means (n)	-0.174*	-0.093*	-0.003	0.031	0.738*	-0.075*	0.094*	0.095*	0.000	75.32
Means (n)	0.151*	0.025*	-0.027*	0.000	-0.130*	0.028*	-0.028*	-0.020*	0.000	75.32

6 Short summary

- The paper studies whether currency risk is priced in U.S. industry expected returns.
- Earlier industry-level work found weak evidence of exchange-rate exposure, even though aggregate market studies found significant currency premia.
- The authors argue that this discrepancy is due to unconditional methodology, not complete hedging.
- They estimate a conditional five-factor model with market, SMB, HML, MAJOR currency, and OITP currency factors.
- Betas are time-varying because covariances and variances are estimated with a multivariate GARCH system.
- Factor returns and exchange-rate changes are predicted using lagged instruments.
- The sample includes 36 U.S. industries from January 1980 to December 1999.
- Simple OLS exposure tests remain weak, matching the old literature.
- The conditional model finds significant currency exposure and significant currency premia for all industries.
- The average absolute currency premium is about 2.47 percentage points per year.
- Currency risk accounts for about 11.7% of total absolute risk premium on average.
- Both developed-country and developing-country currency factors matter, weakening the argument that hedging explains the prior absence of industry currency premia.
- Industry characteristics such as foreign income, competition, liquidity, leverage, and size explain cross-sectional variation in currency premia.
- Macro variables such as trade ratios, growth opportunities, monetary-policy regime, recessions, and currency-crisis dummies explain time variation.
- The broad conclusion is that exchange-rate risk is economically important for U.S. industry cost of equity.

7 Conclusion

7.1 Core conclusion

Francis, Hasan, and Hunter show that currency risk is a priced and economically meaningful component of expected returns for U.S. industries. The average absolute premium is about 2.47 percentage points per year, or about 11.7% of the total absolute risk premium. The weak results in prior industry-level studies are better explained by constant-parameter methods than by complete hedging of currency risk.

7.2 Economic intuition

Exchange-rate movements affect industries through foreign sales, foreign inputs, import competition, foreign customers, and macroeconomic reallocations. These effects change over time, so fixed-coefficient regressions can miss them. When exposures and expected factor returns are allowed to vary, exchange-rate risk becomes visible in the cost of equity.

7.3 Takeaway

The paper reconciles aggregate- and industry-level evidence by showing that industry currency premia are real but conditional. For managers, the result means currency risk can affect the cost of equity even when cash-flow exposure tests look weak. For researchers, the lesson is that hedging is not enough to explain the puzzle; model specification matters.